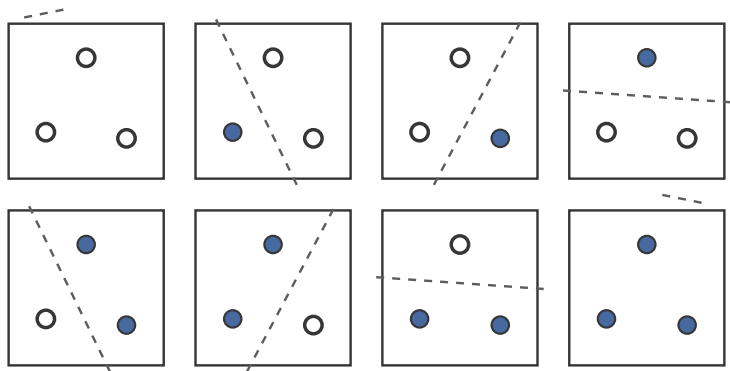


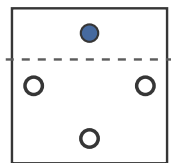
VC dimension via shattering

hypothesis class $\mathcal{H}$ = half-planes (linear classifiers) in $\mathbb{R}^2$

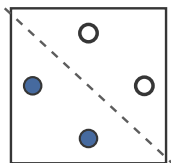
3 points are shattered: all $2^3 = 8$ dichotomies are realizable



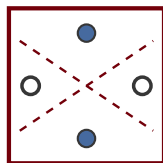
4 points are NOT shattered: the diagonal (XOR) labelling fails



realizable ✓



realizable ✓



not realizable ✗

some 3-set is shattered, no 4-set is $\implies \text{VCdim}(\mathcal{H}) = 3$

● class +1

○ class -1

- - - separating line h